HDOC Day-41 - C Programming

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Problem Statement

Write a menu-driven C program that reads two paragraphs and performs the required string operations using separate user-defined functions.

Operations: Read two paragraphs; find longest and shortest words; remove repeated consecutive spaces; convert paragraphs to title case; find frequency of every distinct word ignoring letter case; and exit.

Header Files Used: stdio.h, string.h, ctype.h

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void readParagraph(char paragraph[], size_t size)
{
    fgets(paragraph, size, stdin);
    paragraph[strcspn(paragraph, "\n")] = '\0';
}

void findLongestAndShortestWords(const char paragraph[],
                                 char longestWord[],
                                 char shortestWord[])
{
    char temp[1000];
    char *word;
    strcpy(temp, paragraph);
    word = strtok(temp, " ,.!?");

    if (word == NULL)
        return;

    strcpy(longestWord, word);
    strcpy(shortestWord, word);

    while (word != NULL)
    {
        if (strlen(word) > strlen(longestWord))
            strcpy(longestWord, word);

        if (strlen(word) < strlen(shortestWord))
            strcpy(shortestWord, word);

        word = strtok(NULL, " ,.!?");
    }
}

void removeRepeatedSpaces(const char paragraph[],
                          char modifiedParagraph[])
{
    int i, j = 0;

    for (i = 0; paragraph[i] != '\0'; i++)
    {
        if (isspace((unsigned char)paragraph[i]))
        {
            if (j > 0 && modifiedParagraph[j - 1] != ' ')
                modifiedParagraph[j++] = ' ';
        }
        else
            modifiedParagraph[j++] = paragraph[i];
    }

    modifiedParagraph[j] = '\0';
```

```
}

void convertToTitleCase(const char paragraph[],
                        char titleCaseParagraph[])
{
    int i;
    int newWord = 1;

    for (i = 0; paragraph[i] != '\0'; i++)
    {
        if (isspace((unsigned char)paragraph[i]))
        {
            titleCaseParagraph[i] = paragraph[i];
            newWord = 1;
        }
        else if (isalpha((unsigned char)paragraph[i]))
        {
            if (newWord)
            {
                titleCaseParagraph[i] =
                    toupper((unsigned char)paragraph[i]);
                newWord = 0;
            }
            else
                titleCaseParagraph[i] =
                    tolower((unsigned char)paragraph[i]);
        }
        else
            titleCaseParagraph[i] = paragraph[i];
    }

    titleCaseParagraph[i] = '\0';
}

void findWordFrequencies(const char paragraph[])
{
    char temp[1000];
    char words[100][50];
    int frequency[100] = {0};
    char *word;
    int count = 0;
    int i, j, found;

    strcpy(temp, paragraph);
    word = strtok(temp, " ,.!?");

    while (word != NULL)
    {
        for (i = 0; word[i] != '\0'; i++)
            word[i] = tolower((unsigned char)word[i]);

        found = 0;

        for (j = 0; j < count; j++)
        {
            if (strcmp(words[j], word) == 0)
            {
                frequency[j]++;
                found = 1;
                break;
            }
        }

        if (!found)
        {
            strcpy(words[count], word);
            frequency[count] = 1;
            count++;
        }

        word = strtok(NULL, " ,.!?");
    }

    printf("\nWord Frequencies:\n");
    for (i = 0; i < count; i++)
        printf("%s : %d\n", words[i], frequency[i]);
}

int main()
{
```

```
    char paragraph1[1000], paragraph2[1000];
    char longest[100], shortest[100];
    char modified[1000], titleCase[1000];
    int choice;

    printf("Enter first paragraph:\n");
    readParagraph(paragraph1, sizeof(paragraph1));

    printf("Enter second paragraph:\n");
    readParagraph(paragraph2, sizeof(paragraph2));

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Find Longest and Shortest Words\n");
        printf("2. Remove Repeated Spaces\n");
        printf("3. Convert to Title Case\n");
        printf("4. Find Word Frequencies\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                findLongestAndShortestWords(
                    paragraph1, longest, shortest);
                printf("\nParagraph 1:\n");
                printf("Longest Word: %s\n", longest);
                printf("Shortest Word: %s\n", shortest);

                findLongestAndShortestWords(
                    paragraph2, longest, shortest);
                printf("\nParagraph 2:\n");
                printf("Longest Word: %s\n", longest);
                printf("Shortest Word: %s\n", shortest);
                break;

            case 2:
                removeRepeatedSpaces(paragraph1, modified);
                printf("\nModified Paragraph 1:\n%s\n", modified);
                removeRepeatedSpaces(paragraph2, modified);
                printf("\nModified Paragraph 2:\n%s\n", modified);
                break;

            case 3:
                convertToTitleCase(paragraph1, titleCase);
                printf("\nTitle Case Paragraph 1:\n%s\n", titleCase);
                convertToTitleCase(paragraph2, titleCase);
                printf("\nTitle Case Paragraph 2:\n%s\n", titleCase);
                break;

            case 4:
                printf("\nParagraph 1");
                findWordFrequencies(paragraph1);
                printf("\nParagraph 2");
                findWordFrequencies(paragraph2);
                break;

            case 5:
                printf("Program exited successfully.\n");
                break;

            default:
                printf("Invalid choice.\n");
        }

    } while (choice != 5);

    return 0;
}
```

Program Output

```
Paragraph 2:
Longest Word: Hello
Shortest Word: bro

----- MENU -----
1. Find Longest and Shortest Words
2. Remove Repeated Spaces
3. Convert to Title Case
4. Find Word Frequencies
5. Exit
Enter your choice: 2

Modified Paragraph 1:
Dhruv pandey

Modified Paragraph 2:
Hello bro

----- MENU -----
1. Find Longest and Shortest Words
2. Remove Repeated Spaces
3. Convert to Title Case
4. Find Word Frequencies
5. Exit
Enter your choice: 3

Title Case Paragraph 1:
Dhruv Pandey

Title Case Paragraph 2:
Hello  Bro

----- MENU -----
1. Find Longest and Shortest Words
2. Remove Repeated Spaces
3. Convert to Title Case
4. Find Word Frequencies
5. Exit
Enter your choice: 4

Paragraph 1
Word Frequencies:
dhruv : 1
pandey : 1

Paragraph 2
Word Frequencies:
hello : 1
bro : 1

----- MENU -----
1. Find Longest and Shortest Words
2. Remove Repeated Spaces
3. Convert to Title Case
4. Find Word Frequencies
5. Exit
Enter your choice: 5
Program exited successfully.
dhruvpandey@DHRUVs-MacBook-Air ~ %
```